STUDENT PERFORMANCE ANALYSIS SYSTEM

Project Presentation

A web-based application for managing student records, assessment marks, and academic performance reports.

PYTHON | STREAMLIT | MYSQL

Presented by: _____ Roll No: _____

1. Project Overview

What the application does

The Student Performance Analysis System is a web application that centralizes academic records and turns assessment data into useful summaries.

Manage records

Store and maintain student details, subject details, and assessment marks.

Analyze performance

Calculate percentages, grades, pass/fail status, and subject-level summaries.

View reports

Explore top performers, students needing support, charts, and downloadable CSV reports.

Main users: academic staff, teachers, or project demonstrators (depending on how the system is deployed).

2. Problem Statement

Why a system like this is useful

Challenges in manual tracking

- Student records may be scattered across files
- Calculating percentages for many assessments takes time
- Identifying students who need support can be difficult
- Producing consistent reports manually is repetitive

Proposed solution

- Centralized student, subject, and marks data
- Automatic calculations and summaries
- Consistent reports and charts
- Faster identification of high and low performance

The system supports academic review; final academic decisions remain with the institution and its staff.

3. Project Objectives

Core goals of the project

01 Maintain student records

Add, edit, and manage student details.

02 Manage subjects

Store subject names, maximum marks, and pass marks.

03 Record assessments

Enter, update, or remove marks for a student and subject.

04 Calculate performance

Generate percentages, grades, and pass/fail indicators.

05 Identify support needs

List students below a selected performance threshold.

06 Create reports

Summarize subject and section performance; export report data.

4. Technologies Used

Tools and their roles

Python

Main programming language for application logic.

Streamlit

Builds the interactive web interface and pages.

MySQL

Stores student, subject, and assessment records.

Pandas

Loads, organizes, groups, and summarizes tabular data.

NumPy

Supports numeric calculations such as percentage handling.

Visual Studio Code

Development environment used to organize and edit project files.

Project files: app.py • database.py • analytics.py • requirements.txt

5. System Architecture

High-level view of how the parts connect



Typical flow

- User interacts with Streamlit pages.
- Python code validates inputs and requests data.
- MySQL stores records and returns query results.
- Pandas and NumPy help prepare summaries for display.

Note: This is a conceptual architecture based on the project files and features.

6. Application Modules

Pages available in the application

Dashboard

High-level overview of academic performance.

Students

Manage student identity, roll number, name, and section.

Subjects

Manage subjects, maximum marks, and pass marks.

Marks

Add assessment marks; update or remove existing entries.

Reports

View student summaries, top performers, support lists, and subject/section analytics.

7. Database Design

Main MySQL tables and relationships

students

- student_id — primary key
- roll_no — unique
- name
- section

subjects

- subject_id — primary key
- subject_name — unique
- max_marks
- pass_marks

marks

- mark_id — primary key
- student_id — foreign key
- subject_id — foreign key
- assessment
- marks_obtained

Important rule: a student cannot have two records with the same subject and assessment name. MySQL enforces this with a unique key on (student_id, subject_id, assessment).

8. System Workflow

From data entry to analysis

1

Add student

Create a student record.

2

Add subject

Set maximum and pass marks.

3

Enter marks

Select student, subject, assessment.

4

Validate

Check inputs and prevent duplicate combinations.

5

Analyze

Calculate percentages, grades, and pass/fail status.

6

Report

View tables, charts, and export CSV.

Data should be entered carefully; duplicate errors usually mean that the same student-subject-assessment combination already exists.

9. Performance Analytics

How the system summarizes marks

Student summary

Total marks, average raw marks, percentage, grade, and overall status.

Subject summary

Assessment count, average marks, average percentage, pass percentage, minimum, maximum, and median.

Grade rules used in analytics.py

Percentage	Grade
90% and above	A
80%–89.99%	B
70%–79.99%	C
60%–69.99%	D; below 60% = F

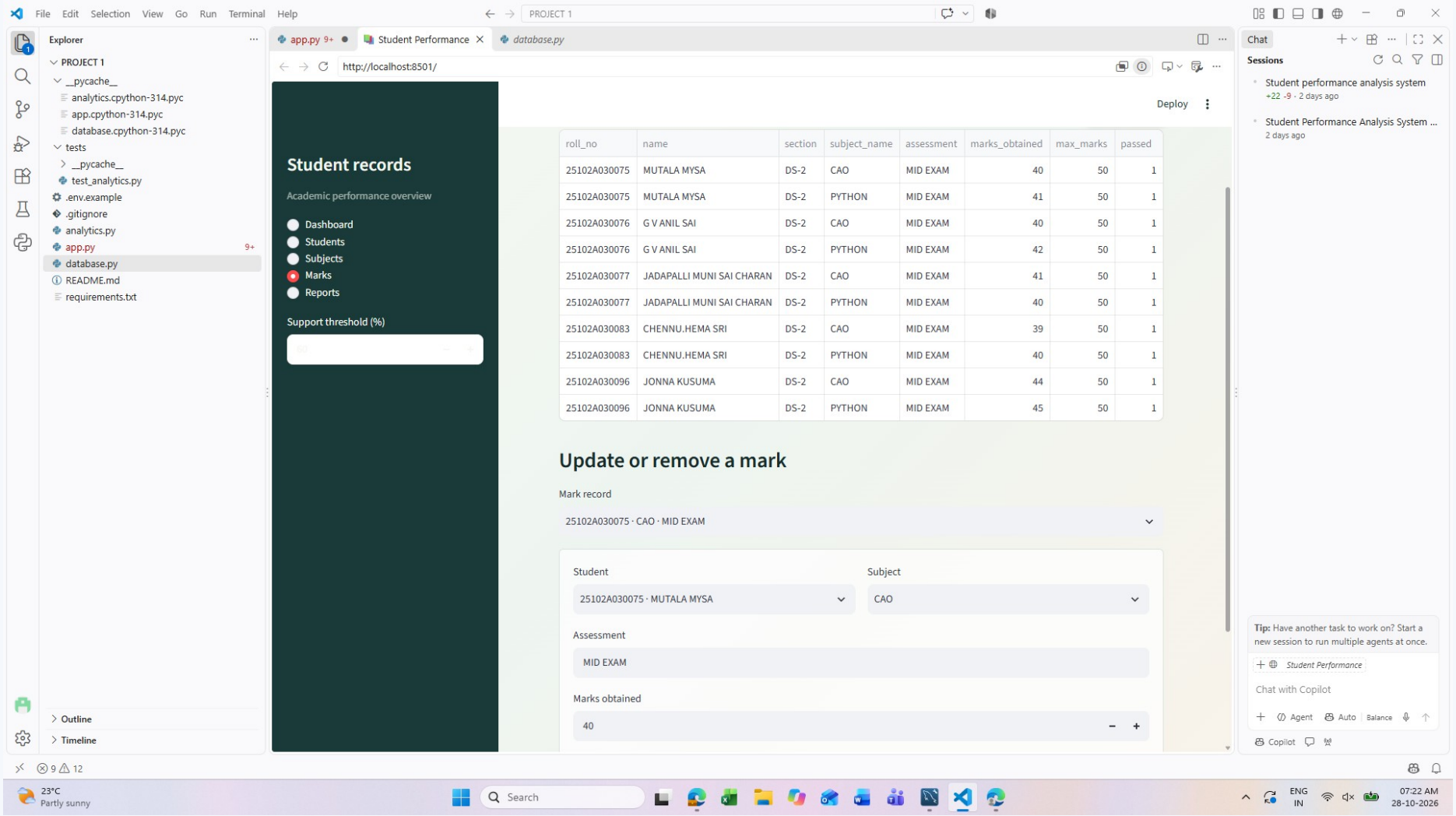
Pass / fail interpretation

A student is marked Pass only when every recorded assessment meets that subject's pass-mark requirement. Otherwise the student is marked Fail.

Percentages are normalized against maximum marks. Average raw marks may be less comparable when subjects have different maximum marks.

10. Application Screenshot

Marks page: existing records and update form



The screenshot shows the Marks module and the “Update or remove a mark” section.

11. Testing and Validation

Example test cases for the project

Test	Action	Expected result
TC-01	Add a student with a new roll number	Student record is saved
TC-02	Add a duplicate roll number	System rejects duplicate record
TC-03	Enter marks within valid range	Assessment is saved
TC-04	Enter same student + subject + assessment again	Duplicate constraint prevents duplicate
TC-05	Update marks for an existing record	Record is updated if no conflict
TC-06	Open Reports page	Summary tables and charts are displayed
TC-07	Download report CSV	CSV file is downloaded

Replace or supplement these example cases with the actual results from your own testing.

12. Benefits and Future Scope

What works now and what could be improved

Benefits

- Centralized records
- Faster calculations
- Clear performance summaries
- CSV export for further analysis

Current considerations

- Accuracy depends on entered data
- Duplicate assessment combinations are restricted
- Access control and deployment settings should be reviewed before real-world use

Future enhancements

- Login and role-based access
- PDF report generation
- Attendance integration
- Trend analysis across terms
- Automated backup

Future enhancements are suggestions, not features claimed as already implemented.

CONCLUSION

The Student Performance Analysis System combines student and subject management with assessment tracking and performance analytics.

It helps present academic data in a structured form through summaries, reports, and charts. The project can be extended with secure user accounts, additional reporting formats, and longitudinal analysis.

THANK YOU

Questions?